METRIC

A-A-59139A(OS)
April 2, 2024
SUPERSEDING
A-A-59139
September 25, 1998

COMMERCIAL ITEM DESCRIPTION

ALUMINUM POWDER, ATOMIZED (FOR USE IN EXPLOSIVES)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers one type of atomized aluminum powder for use in explosives.

2. SALIENT CHARACTERISTICS.

2.1 Material. The aluminum powder shall be microfine virgin aluminum particles manufactured by an atomizing process. The shape of the particles shall be spheroidal.

2.2 Aluminum purity. The aluminum purity shall be a minimum of 98.5 percent when determined by difference with the calculated allowable impurities specified in 2.2.1 through 2.2.5.

2.2.1 Elemental impurities. The elemental impurities (silicon, iron, zinc, and copper) shall not exceed the maximums specified in table I when determined by inductively coupled plasma atomic emission spectrometry testing in accordance with ASTM E3061.

TABLE I. Maximum allowable impurities.

Impurity	Maximum percent allowable	Verification
Silicon	0.3	2.2.1
Iron	0.5	2.2.1
Zinc	0.05	2.2.1
Copper	0.1	2.2.1
Grit	0.05	2.2.2
Oil and grease	0.2	2.2.3
Volatile matter at 105 °C	0.1	2.2.4
Alkalinity as magnesium hydroxide	0.07	2.2.5

2.2.2 Grit. The grit shall not exceed the maximum specified in table I when determined in accordance with the following procedure:

Weigh 5.0 to 5.1 grams of aluminum powder to the nearest 0.01 gram and place it into a 400-milliliter (mL) beaker. Cover with approximately 10 mL of water. Cautiously and slowly, add 60 mL of concentrated hydrochloric acid. Cover with a ribbed watch glass, heat to effect solution of the sample, and cool. Then, transfer sufficient solution to a tared 50-mL centrifuge tube to balance a second centrifuge tube containing approximately 45 mL of water.

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FSC 6810

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Centrifuge the two centrifuge tubes, in balanced position, at 600 to 700 relative centrifugal force (see 6.3) for 15 minutes. Allow the centrifuge to stop without braking and decant the liquid layer from the centrifuge tube containing the solution. Add the rest of the solution to the centrifuge tube. Rinse the beaker with water and add the rinsing to the solution centrifuge tube. Add a sufficient quantity of rinsing water to the specimen tube to balance it with the centrifuge tube containing water. Centrifuge, stop, and decant as above. Add sufficient water to balance the centrifuge tube, shake to mix, and centrifuge as above. Decant the liquid layer and dry the tared centrifuge tube to a constant weight in a drying oven at 105±2 °C. Cool in a desiccator and weigh to the nearest 0.1 milligram. Calculate percent grit as follows:

$$\text{Percent grit} = \frac{100(A - B)}{W}$$

Where:

A = Weight of centrifuge tube and dried contents in grams

B = Tare weight of centrifuge tube in grams

W = Weight of specimen in grams

2.2.3 Oil and grease. The percent of easily extracted fatty and oily matter shall not exceed the maximum specified in table I when determined in accordance with the procedure for aluminum in ASTM D480.

2.2.4 Volatile matter at 105 °C. The percent of volatile matter shall not exceed the maximum specified in table I when determined in accordance with the procedure for aluminum paste in ASTM D480, except that a 10-gram specimen shall be used. Calculate the percent of volatile matter as follows:

$$\text{Volatile matter at 105 °C (percent)} = \frac{\text{Weight Loss}}{\text{Weight of Sample}} \times 100$$

2.2.5 Alkalinity as magnesium hydroxide. The alkalinity as percent magnesium hydroxide shall not exceed the maximum specified in table I when determined in accordance with the following procedure:

Weigh approximately 2 grams of test sample to the nearest milligram and transfer to a stoppered Erlenmeyer flask. Add 200 mL of cold water. Stopper and shake the flask every 2 minutes. After 15 minutes, filter the sample through a dry, neutral filter paper. Titrate 100 mL of filtrate with 0.05N sulfuric or hydrochloric acid using five drops of bromothymol blue as the indicator. Run a blank determination. Calculate alkalinity as percent magnesium hydroxide as follows:

$$\text{Percent alkalinity} = \frac{5.833N\,(A - B)}{W}$$

Where:

A = mL of acid required for sample titer

B = mL of acid required for blank titer

N = Normality of acid

W = Weight of sample in grams

2.3 Screen analysis. The material shall conform to the following requirements for the screen analysis:

a. A minimum of 95.0 percent shall pass through a U.S. No. 200 sieve.

b. A minimum of 80.0 percent shall pass through a U.S. No. 325 sieve.

The particle size distribution of aluminum powder shall be determined in accordance with ASTM B214 using U.S. Standard sieve sizes in accordance with ASTM E11 and 50-gram samples. A weighed portion of approximately 50 grams of the sample shall be placed on the top sieve of the nest of sieves assembled as follows, top to bottom: 100, 200, 230, 325, and bottom pan.

2.4 Average particle size. The average particle size of the material shall be 11 to 27 micrometers as determined by a sub-sieve sizer (see 6.4) according to the manufacturer's instructions.

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3. REGULATORY REQUIREMENTS. This standard does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of whoever uses this standard to consult and establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.2 Inspection requirements. Ten percent of each single batch or continuous production run (produced using a single set of operations and operating conditions) shall be randomly selected for acceptance tests. The sample shall be uniformly mixed, and an 8-ounce composite sample shall be selected for the tests.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 ASTM standards. ASTM standards are available online at www.astm.org.

6.1.2 Federal standards. Federal standards are available online at https://quicksearch.dla.mil.

6.2 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this commercial item description.

b. Packaging and packing requirements.

c. Activities requiring copies of completed Safety Data Sheets (SDS) (see 6.5).

6.3 Relative centrifugal force (rcf). Relative centrifugal force is defined as follows:

$$\text{rcf} = A \times B^2 \times 1.118 \times 10^{-5}$$

Where:

A = Rotational radius in centimeters

B = Speed of centrifuge in revolutions per minute

6.4 Sub-sieve sizer. The sub-sieve sizer used to measure the average particle size is available from Fisher Scientific (CAGE: 5RZ49), www.fishersci.com, +1 (800) 766-7000, 168 Third Ave., Waltham, MA 02451, product MDL95; or may be available from equivalent suppliers.

6.5 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 should be in the contract.

6.6 Key words.

Energetic ingredient

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodian:
Navy – OS

Review activity:
DLA – GS

Preparing activity:
Navy – OS
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